Fiscal Year:	FY 2010	**Task Last Updated:**	FY 12/13/2010
PI Name:	Smith, Scott M Ph.D.		
Project Title:	Vitamin D supplement evaluation of dosing requirements		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Food and Nutrition**:Risk of Performance Decrement and Crew Illness Due to Inadequate Food and Nutrition (2) **Nutrition**:Risk of Inadequate Nutrition		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	NASA Johnson Space Center		
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City:	Houston	**State:**	TX
Zip Code:	77058-3607	**Congressional District:**	36
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	10/01/2007	**End Date:**	09/30/2010
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Locke, Jim (NASA Johnson Space Center) Zwart, Sara (USRA/NASA Johnson Space Center) Parsons, Howard (Canadian Space Agency) Kimlin, Michael (Queensland Univ Technology, Brisbane, Australia)		
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Risk of vitamin D insufficiency or deficiency is high for space travelers because they lack ultraviolet light exposure and have an insufficient dietary supply of vitamin D. Despite the provision of vitamin D supplements to International Space Station (ISS) crewmembers, their vitamin D status is consistently lower after flight than before flight, and in several crewmembers it has decreased to levels considered clinically significant. Vitamin D has long been known to play a role in calcium metabolism, and more recently its non-calcitropic functions have been recognized. According to the results of several recent studies, functionally relevant measures indicate that the lower limit of serum 25 hydroxyvitamin D (the key marker of vitamin D status) should be raised from the current 23 nmol/L to 80 nmol/L. The mean preflight serum 25-hydroxyvitamin D for U.S. ISS crewmembers to date is 63 ± 16 nmol/L, and after a 4- to 6-mo space flight it typically decreases 25 to 30 percent despite provision of supplements (400 IU/d). The sub-optimal pre- and postflight vitamin D status is an issue that needs to be addressed, and part of addressing this is defining the amount of supplemental vitamin D required to reach and maintain optimal vitamin D status in space flight crewmembers. This is very important for crewmembers on long duration missions, and is critical for exploration-class missions. We recently completed a ground-based investigation evaluating the efficacy of vitamin D supplementation during the winter months in Antarctica, when UV-B radiation levels are essentially zero. A supplement of 2,000 IU/d raised serum 25-hydroxyvitamin D to acceptable levels, but as with any time a daily supplement or change in routine is required, compliance can be an issue. In this dosing study, we are investigating whether a weekly or monthly dose of vitamin D could be substituted for a daily dose. We are monitoring clinical markers in real time for any sign of vitamin D toxicity. Data from this study will help us to provide long-duration space flight crewmembers an evidence-based recommendation for safe and effective vitamin D supplementation to achieve and maintain optimal vitamin D status before, during, and after flight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	In this study, we investigated whether a weekly or monthly dose of vitamin D could be substituted for a daily dose with respect to effectiveness. Data from this study will help us to provide long-duration space flight crewmembers a safe, effective, and evidence-based recommendation for vitamin D supplementation to achieve and maintain optimal vitamin D status before, during, and after flight. This is very important for long duration crewmembers, and is critical for exploration-class missions. The results will also inform the general scientific and medical communities, where vitamin D supplementation has become an ongoing topic of discussion and debate.
Task Progress:	The data have been analyzed, and submitted for peer-review/publication (submitted in December 2010). Briefly, the data provide evidence that 2000 IU daily, 10,000 IU once weekly, or 50,000 IU (weekly then) monthly dosing regimens used in this study were similarly effective at raising circulating vitamin D status over the course of 90 days.
Bibliography Type:	Description: (Last Updated: 01/19/2026)
Articles in Peer-reviewed Journals	Zwart SR, Parsons H, Kimlin M, Innis S, Locke J, Smith SM. "Acute and chronic effects of 3 dosing regimens of vitamin D supplementation." American Journal of Clinical Nutrition, submitted, December 2010. , Dec-2010
Articles in Peer-reviewed Journals	Smith SM, Gardner KK, Locke J, Zwart SR. "Vitamin D supplementation during Antarctic winter." Am J Clin Nutr. 2009 Apr;89(4):1092-8. Epub 2009 Feb 18. PMID: 19225122 , Apr-2009